

SHUVAM PAL

+91 85830 54679 ✉ ishuvam.pal@gmail.com  Shuvam Pal  unichronic

Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Product Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built production workflow software with API lookups, browser automation, validation checkpoints, retry handling, and human-reviewable state for high-friction filings.
- Reduced manual filing time by **85%** by turning repeated operational steps into recoverable backend workflows.
- Used Docker, Google Cloud Run, and GitHub Actions to keep service deployment and iteration predictable.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Integrated Python model-output workflows into medical-imaging tooling, covering preprocessing, segmentation masks, label mapping, large MRI volumes, and visual inspection.
- Reduced runtime by **87%** through asynchronous and parallel execution while keeping outputs traceable through generated masks and 3D surfaces.
- Worked through correctness issues across NumPy data handling, PyTorch/ONNX-style inference, VTK rendering paths, and wxPython UI integration.

Open Source

[Contributions](#) 

- Contributed to research-adjacent open source including **Invesalious** medical-imaging workflows, **IOOS** scientific-data tooling, and **Tiled** visualization code, plus compatibility and security-validation fixes.

Projects

Hyoka | *Python, FastAPI, PostgreSQL, OpenTelemetry, SQLAlchemy, Typer, Docker*

- Built an **AI-agent reliability control plane** with SDK/proxy/OTLP trace ingestion, evaluator registries, replay modes, failure mining, validation runs, release gates, promotions, and audit logs.
- Implemented worker-owned leases, content-addressed artifacts, signed manifests, project API keys, SQLAlchemy/Alembic models, and reproducible runs around immutable inputs.

Seaweed: Contest Platform | *Go, TypeScript, React, PostgreSQL, Redis, Judge0, Docker, Kubernetes*

[GitHub](#) 

- Built product and platform paths for registration, submissions, sandboxed code execution, admin review, live rankings, and durable result records across contest traffic.
- Designed async verdict polling, PostgreSQL result schemas, Redis-backed runtime state, Docker/Kubernetes deployment, and **sub-5s P95 verdict latency**.

Swish Support System | *Python, FastAPI, React, PostgreSQL/pgvector, Redis, Langfuse*

[GitHub](#) 

- Built a hybrid AI + rules support workflow where LLM assessments classify ambiguous complaints while deterministic policy code controls refunds, replacements, coupons, and escalations.
- Grounded responses in order, item, evidence, customer, and policy context with pgvector retrieval, Redis state, Langfuse traces, confidence thresholds, and regression cases.

Technical Skills

Languages: Python, TypeScript, Go, SQL, C++

Frameworks: PyTorch, FastAPI, React, Next.js, VTK, wxPython, Echo

Tools: NumPy, ONNX, PostgreSQL, Redis, Docker, GCP, GitHub Actions, OpenTelemetry, Prometheus, Judge0